

SCIENTIFIC AND TECHNOLOGICAL CAPABILITIES AND ECONOMIC CATCH-UP*

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This paper seeks to explain why the Philippines has remained technologically and economically underdeveloped up to now and what it will take for the country to catch up scientifically, technologically, and economically. The paper first reviews the extent to which the Philippines has been left behind in S&T and economic development. Next it discusses and confutes two approaches to the problem of economic and S&T under-development that have been pushed in the past: (1) the "science-push" approach advocated by basic scientists starting with Vannevar Bush and (2) the "market-pull" approach favored by mainstream (neoclassical, neoliberal) economists and traditional businessmen and based on the neoclassical economic theory of comparative advantage. The paper then presents an alternative approach — a "catch-up oriented", "capability-based", technonationalist approach — which draws from the successful catch-up experiences of East Asian NICs. The approach entails the abandonment of the precepts of neoliberalism and the Washington Consensus, seeks to achieve rapid national economic and S&T catch-up, and calls for the integrated upgrading of the supply, demand, and linkage parts of the national S&T system to world-class standards.

Keywords: Technological and economic catch-up, technological capabilities, technonationalism, technoliberalism

I. INTRODUCTION

It has become a consensus that the Philippines was second only to Japan in East Asia in industrial, technological, and economic development in the 1950s. Since then, however, the Philippines has been overtaken by its neighbors in East Asia with the exception of Vietnam, Myanmar, Laos, and Cambodia.

So why has the Philippines been left behind by its East Asian neighbors? Why has the country remained underdeveloped scientifically, technologically, and economically? Why has it failed to become a

newly industrialized country up to now? Is it because the country neglected to develop its science and technology or S&T? Or is it because the Philippines adopted protectionist, nationalist policies for too long and failed to adopt open, liberal policies soon enough?

This paper tries to explain why the Philippines has remained a scientific, technological, and economic laggard up to now and what needs to be done to enable the country to catch up scientifically, technologically, and economically. Section II first gives a brief overview of the extent to

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